

HYERIN CHO (조혜린)

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[Website](#) ◇ [Google Scholar](#) ◇ [Github](#)

RESEARCH INTERESTS

High-energy astrophysics; Black holes; General relativity; Plasma physics; Galaxy evolution; GRMHD simulations; Accretion; Relativistic Jets

EDUCATION & APPOINTMENTS

Caltech Burke Institute Fellow & CITA (UofT) Fellow Sep. 2026 - present

Harvard University, USA May 2026

Ph.D. in Astronomy and Astrophysics

Advisor: Ramesh Narayan

GIST(Gwangju Institute of Science and Technology), Korea Feb. 2020

B.S. Physics Major / Math Minor (*cum laude*)

PUBLICATIONS

h-index: 11, citations > 700 (10/04/2026) [Google Scholar](#), [ads](#), [arXiv](#)

First-author articles

1. **Cho H.**, Prather C., Narayan R., Su K.-Y., Ricarte A., Natarajan P., Porras-Valverde A. J., 2026, [ApJL](#), “*Bridging Scales in Black Hole Accretion and Feedback: Subgrid Prescription from First Principles*”
2. **Cho H.**, Prather B. S., Narayan R., Su K.-Y., Natarajan P., 2025, [ApJ](#), “*Bridging Scales in Black Hole Accretion and Feedback: Relativistic Jet linking the Horizon to the Host Galaxy*”
3. **Cho H.** and Narayan R., 2025, [ApJ](#), 991, “*Variability in Black Hole Accretion: Dependence on Rotational and Magnetic Energy Balance*”
4. **Cho H.**, Prather B. S., Su K.-Y., Narayan R., Natarajan P., 2024, [ApJ](#), 977, “*Multizone Modeling of Black Hole Accretion and Feedback in 3D GRMHD: Bridging Vast Spatial and Temporal Scales*”
5. **Cho H.**, Prather B. S., Narayan R., Natarajan P., Su K.-Y., Ricarte A., Chatterjee K., 2023, [ApJL](#), 959, “*Bridging Scales in Black Hole Accretion and Feedback: Magnetized Bondi Accretion in 3D GRMHD*”
 - featured in [AASnova](#)
6. **Cho H.** and Narayan R., 2022, [ApJ](#), 932, “*Analytical Model of Disk Evaporation and State Transitions in Accreting Black Holes*”
7. **Cho H.**, Macquart J.-P., Shannon R. M., Deller A. T., Morrison I. S., Ekers R. D., Bannister K. W., et al., 2020, [ApJL](#), 891, “*Spectropolarimetric analysis of FRB 181112 at microsecond resolution: Implications for Fast Radio Burst emission mechanism*”
 - featured in [astrobites](#)

PRESENTATIONS

Seminars & Colloquia & Meetings († invited)

1. † Accretion-Ejection Seminar (online) Apr. 2026
2. ITC luncheon talks (Harvard, USA) Apr. 2026
3. † BHEX AGN demographics meeting (online) Mar. 2026
4. † Kavli Institute for Particle Astrophysics and Cosmology (KIPAC) Tea Talk (online) Nov. 2025

5. † Theoretical High Energy Astrophysics Group meeting (Columbia, USA)	Nov. 2025
6. † CCA Stars & Plasma Astrophysics group meeting (CCA, USA)	Nov. 2025
7. † Bahcall lunch talks (IAS, USA)	Nov. 2025
8. † TAPIR seminar / Astronomy Tea Talks (Caltech, USA)	Oct. 2025
9. † CIERA Theory Group Meeting (Northwestern, USA)	Oct. 2025
10. † CCA-NY area community galaxy formation seminar (CCA, USA)	Oct. 2025
11. † EHT theory working group (online)	Sep. 2025
12. Umbrella Dialogues (Harvard, USA)	Jan. 2025
13. † Quataert group meeting (Princeton, USA)	Nov. 2024
14. † Astro coffee (IAS, USA)	Nov. 2024
15. † Strong Gravity Seminar (Perimeter Institute, Canada)	Apr. 2024
16. † Plasma-astro meeting (CITA, Canada)	Apr. 2024
17. † University of Illinois Urbana-Champaign Journal Club (online)	Feb. 2024
18. ITC luncheon talks (Harvard, USA)	Feb. 2024
19. Kim group meeting (Seoul National University, Korea)	Jan. 2024
20. † ITC luncheon talks (Harvard, USA)	Mar. 2023
21. † ZARM Accretion disk and relativistic astrophysics (online)	Nov. 2022
22. † Curtin University Colloquium (online)	May 2022

Conference & Workshop († invited)

1. Talk AGN & Energy Flows Workshop (CCA, USA)	2025
<i>“Bridging Scales in Black Hole Accretion and Feedback via the Multizone method”</i>	
2. † Talk The multiscale environment of AGN across cosmic time (MPA, Germany)	2025
<i>“Bridging Scales in Black Hole Accretion and Feedback: Relativistic Jet linking the Horizon to the Host Galaxy”</i>	
3. Talk Deciphering Cosmic Code for Galaxy Formation (Puerto Varas, Chile)	2024
<i>“Bridging Scales in Black Hole Accretion and Feedback in GRMHD”</i>	
4. Talk Tinsley Workshop (Yale, USA)	2024
<i>“Bridging Scales in the Black Hole Accretion-Feedback”</i>	
5. Talk Bridging Scales Workshop (Harvard, USA)	2024
<i>“Simulations of Coupled Feeding & Feedback”</i>	
6. Talk The Event Horizon and Beyond - Celebrating 50 Years of Narayan (Harvard, USA)	2024
<i>“Bridging Scales in AGN Accretion and Feedback”</i>	
7. Talk Resolving Galaxy Ecosystems Across All Scales (CUHK, Hong Kong)	2023
<i>“Bridging scales between supermassive black holes and their host galaxies via multi-annuli approach”</i>	
8. Talk Black Holes on Broadway: The Next Generation of AGN Models in Galaxy Formation (CCA, USA)	2023
<i>“Bridging Scales between Supermassive Black Holes and Their Host Galaxies via Multi-annuli Approach”</i>	
9. Talk Boston-Area Blackhole Accretion Meeting (BABAM!) (Harvard, USA)	2023
<i>“Bridging Scales in Black Hole Accretion and Feedback”</i>	
10. Poster AAS (Pasadena, USA)	2022
<i>“Analytical Model of Disk Evaporation and State Transitions in Accreting Black Holes”</i>	
11. Talk ngEHT meeting (Granada, Spain)	2022
<i>“Analytical Model of Disk Evaporation and State Transitions in Accreting Black Holes”</i>	
12. Talk New England Regional Quasar and AGN Meeting (NERQUAM) (UConn, USA)	2022
<i>“Analytical Model of Disk Evaporation and State Transitions in Accreting Black Holes”</i>	

AWARDS & FELLOWSHIPS

Flatiron Research Fellowship (FRF) , Center for Computational Astrophysics (declined)	2026
Princeton-IAS Joint Fellowship (declined)	2026
Kavli Institute Fellowship , Cambridge University (declined)	2026
KIPAC Bunyan Fellowship , Stanford University (declined)	2026
CIERA Fellowship , Northwestern University (declined)	2026
ACCESS Explore (PI: Hyerin Cho , PHY250303)	2025 - present
ACCESS Maximize (PI: Ramesh Narayan, AST080028)	2024 - 2025
ACCESS Explore (PI: Hyerin Cho , PHY230079)	2023 - 2024
Ilju Foundation Study Abroad Scholarship Selected as one of 6 recipients nationwide in Korea across all disciplines for a prestigious doctoral scholarship (~ \$120,000 USD over four years) awarded to support Ph.D. studies abroad.	2020 - 2024
Talent Award of Korea (대한민국인재상) Awarded by the South Korean government to outstanding young leaders demonstrating excellence, creativity, and public service. Selected as one of the top national recipients representing Korea's next-generation talent.	2020
GIST Outstanding Thesis Award (우수논문상)	2020
GIST Future Research Talent Award (미래인재상)	2020
Korea National Science and Engineering Scholarship (국가이공계장학금) A scholarship to fund full tuition for 8 semesters from Korea Student Aid Foundation, Ministry of Education	2015 - 2020

TEACHING EXPERIENCE

Harvard University	
Teaching Assistant Galactic and Extragalactic Astronomy (undergraduate course)	Fall 2022
Teaching Assistant Radiative Processes in Astrophysics (graduate course)	Fall 2021
GIST	
Teaching Assistant Introduction to Topology (4th-year course)	Fall 2019
Teaching Assistant Electromagnetism II (3rd-year course)	Spring 2018

SERVICE, OUTREACH, ORGANIZATIONAL EXPERIENCE

Outreach	
• Black Hole Scholars talk at Valle Verde Early College High School	2025
• Astronomy on Tap talk	2025
• Crescent Street Halloween Party “Ask an Astrophysicist”	2025
• Research Talk for Harvard-Smithsonian Science Research Mentoring Program (SRMP)	2024
• Black Hole Scholars talk at Buckingham Browne & Nichols School	2024
• Interviewed as a guest of an episode of Astrophiz podcast	2023
• Translating articles in Korean for World of Women in STEM (WOWSTEM) website	2022 -
• Women in STEM (WiSTEM) Mentor	2020 - 2021
Peer Review: Nature astronomy	2021
Conference Services	

- **Local Organizing Committee** [RN50 Conference](#) The Event Horizon and Beyond - Celebrating 50 Years of Narayan 2024
- **Session Chair** 7th Annual [Black Hole Initiative Conference](#) 2024
- **Local Organizing Committee** [Bridging Scales Workshop](#) 2024
- **Local Organizing Committee** Boston-Area Blackhole Accretion Meeting (BABAM!) 2023

Narayan Group Meeting Organizer 2023 - 2026

REFERENCES

Ramesh Narayan

Thomas Dudley Cabot Professor of the Natural Sciences
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Priyamvada Natarajan

Joseph S. and Sophia S. Fruton Professor of Astronomy and Professor of Physics
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